

BEYOND SEGMENTATION: CONTROLS TO KEEP ON YOUR RADAR

Segmentation, QoS and VPNs are today's essentials, but the threat landscape – and India's rule-book – moves fast. Here are six emerging measures worth weaving into next quarter's roadmap.

1. Make the Jump to WPA3-Enterprise

WPA3 brings SAE handshake hardening for personal networks and 192-bit cryptography for enterprise mode, shutting down brute-force attacks that still work against WPA2 PSK. The Wi-Fi Alliance recommends SMBs adopt WPA3 on all new APs shipped since 2023, many of which can enable it with a firmware toggle.

2. Swap Shared Secrets for Certificates (EAP-TLS)

Rolling Wi-Fi passwords every month is labour-intensive and error-prone. Certificate-based 802.1X authentication (EAP-TLS) binds each device to its own X.509 cert, enabling revocation without touching anyone else's access. Open-source PKI stacks such as Smallstep let you automate issuance in under an hour.

3. Encrypt Your DNS, Too

Even inside a VPN, split-tunnelled users may leak lookups. DNS-over-TLS and DNS-over-HTTPS wrap queries in TLS, blocking common phishing redirects and middle-box snooping. Choose DoH for easy browser support or DoT for simpler firewall whitelisting – both beat port-53 in the clear.

4. Align with ISO 27001 Annex A 8.22 – Segregation of Networks

Many investors now ask Indian startups for ISO 27001 attestation before signing term-sheets. Clause 8.22 explicitly requires traffic controls “based on business requirements,” so the VLAN and ACL designs

from Chapter 2 map neatly to audit check-boxes and speed certification.

5. Understand the New Cost of Complacency – Cyber-Insurance Premiums

Insurers raised rates an average 18 % last year for firms lacking MFA or robust segmentation. Underwriters increasingly demand proof of least-privilege network design, real-time logging and rapid patching before quoting. Building these controls now can shave lakhs off next renewal.

6. Remote-Work Compliance After India's VPN Data-Retention Rule

Since 2022, VPN providers operating Indian servers must log user names, IP addresses and timestamps for five years. Some global services have shut domestic nodes, meaning your IPsec tunnels may exit via Singapore or Dubai – check latency budgets and update policies to reflect offshore routing.

Bonus Trend: SASE & Zero Trust Network Access for SMBs

Secure Access Service Edge bundles cloud firewalling, CASB and ZTNA into a single subscription – appealing when headcount outpaces rack space. Analysts predict a double-digit jump in Indian SMB adoption through 2025 as bandwidth-based pricing falls.

Why It Matters

Implementing just two of these controls – certificate-based Wi-Fi and encrypted DNS – cuts credential theft and spoofed-DNS incidents by over 60 % in small-office studies. Layer in ISO-aligned segmentation and you not only please auditors but also trim cyber-insurance premiums, all while staying on the right side of India's evolving data laws and remote-work realities.

6) Pulling it all together

- **VLANs + ACLs** wall off threats before they pivot.
- **QoS policies** keep Zoom clear and payroll timely.
- **Captive portals** offer hospitable Wi-Fi without exposing assets.
- **VPN bridges** extend zero-trust boundaries to every living room in India's Tier-III towns.

- **Policy templates** hard-code good hygiene – no exceptions for that “just one file” USB stick.

Print the cheat-sheets, paste them near the switch rack, and share the scripts with your neighbourhood IT WhatsApp group. Securing a modern office network isn't rocket science – it's disciplined slicing, shaping, and safeguarding, one VLAN tag and OTP at a time. **d**